Örnek 1:

$4^3 = 4.4.4 = 64$

$(-2)^4 = (-2).(-2).(-2).(-2) = 16$

$(-5)^3 = (-5).(-5).(-5) = -125$

Örnek 2:

$29^0 = 1$

$\left(-\frac{2}{5}\right)^{-1} = \left(-\frac{5}{2}\right)$

$(9)^0 = 1$

$\left(\frac{5}{4}\right)^{-2} = \left(\frac{4}{3}\right)^2 = \frac{16}{9}$

$-2014^0 = -1$

$(-2)^{-5} = \left(-\frac{1}{2}\right)^5 = -\frac{1}{32}$

Örnek 3:

$-\left[\left(-\frac{1}{2}\right)^{-1}\right]^4 = -\left(-\frac{1}{2}\right)^{-4} = -(-2)^4 = -2^4 = -16$

Örnek 4:

$-4^5 + 3 4^5 - 7.3^4 - 3.3.4 - 4^5 = 4^5.(-1+3-1) + 3^4.(-7-3)$

Örnek 5:

$\frac{5^{80} + 5^{85} + 5^{90}}{5^{85} + 5^{80} + 5^{75}} = x$ ise $\frac{x}{125} = ?$

$\frac{5^{80}(1 + 5^5 + 5^{10})}{5^{75}(5^{10} + 5^5 + 1)} = 5^5$ $\qquad \frac{x}{125} = \frac{5^5}{5^3} = 5^{5-3} = 5^2 = 25$

Örnek 6:

$\frac{14^n + 6^n + 2^n}{7^n + 3^n + 1} = ?$ $\qquad \frac{2^n.(7^n + 3^n + 1)}{7^n + 3^{n+1}} = 2^n$

Örnek 7:

$(0,027)^{\frac{5}{3}} . 10^5 = ?$

$(27.10^{-3})^{\frac{5}{3}} . 10^5 = (3^3)^{\frac{5}{3}} . (10^{-3})^{\frac{5}{3}} . 10^5 = 3^5 . 10^0 = 3^5$

Örnek 8:

$$\frac{-9^{-2}.25^{-1}}{125^{-1}.27^{11}} = \frac{-3^{-4}.5^{-2}}{5^{-3}.3^{-3}} = \frac{5^{3-2}}{3^{4-3}} = -\frac{5}{3}$$

Örnek 9:

$$\frac{x^{a+1}}{x^{a-2}} : \left(\frac{x^{a+2}.x^{a+5}}{x}\right)^{-1} = x^{3}.x^{-a-4} = x^{-a-1}$$

Örnek 10:

$$\left(\frac{a^x}{a^y}\right)^{x-y}.\left(\frac{a^y}{a^x}\right)^{x-y} = (a^{x-y+y-x}) = (a^0)^{x-y} = 1^{x-}$$

Örnek 11:

$3^x = 2 \quad 27^{x+1} - 81^x = ?$

$$(3^x)^3.3^3 - (3^x)^4 = 2^3.3^3 - 2^4 = 200$$

Örnek 12:

$x-2=1 \quad |x-1|+|1-x| = ?$

$x-1=2 \quad y=x=-2 \quad |2|+|-2| = 2+2=4$

Örnek 13:

$x \in R$ ve $-2<x<4 \quad |x+2|+|x-4| = ?$

$x+2>0 \quad x-4<0 \quad x+2-(x-4) = x+2-x+4 = 6$

Örnek 14:

$x<0<y \quad \left|\frac{2x-y}{x+y}\right| - |2y-x| = ?$

$2x-1<0 \quad 2y-x>0 \quad \frac{-(2x-1)-(2y-x)}{x+y} = -\frac{(x+1)}{x+y} = -1$

Örnek 15:

$\sqrt{25} \in \mathbb{N}$ $\sqrt{0} \in \mathbb{R}$ $\sqrt[4]{81} \in \mathbb{R}$, $\sqrt{-3} \in \mathbb{R}$

Örnek 16:

$\sqrt{0,64} + \sqrt{0,04} - \sqrt[3]{0,027}$

$\sqrt{\left(\frac{8}{10}\right)^2} + \sqrt{\left(\frac{2}{10}\right)^2} - \sqrt[3]{\left(\frac{3}{10}\right)^3} = \frac{8}{10} + \frac{2}{10} - \frac{3}{10} = \frac{7}{10}$

Örnek 17:

$\frac{\sqrt{75} - \sqrt{27} + \sqrt{12}}{\sqrt{108} + \sqrt{48}} = \frac{4\sqrt{3}}{10\sqrt{3}} = \frac{2}{5}$

Örnek 18:

$\sqrt{2}.(\sqrt{18} + \sqrt{50} - \sqrt{32} + \sqrt{8})$

$\sqrt{36} + \sqrt{100} - \sqrt{64} + \sqrt{16} = 6 + 10 - 8 + 4 = 12$

Örnek 19:

$\frac{\sqrt[5]{16}}{\sqrt[5]{\frac{1}{2}}} = \sqrt[5]{\frac{16}{\frac{1}{2}}} = \sqrt[5]{32} = 2$

Örnek 20:

$\sqrt{25 - 25x^2} - \sqrt{64 - 64x^2} = 5\sqrt{1-x^2} - 8\sqrt{1-x^2} = -3\sqrt{1-x^2}$

Örnek 21:

$\sqrt{2\sqrt{3\sqrt{3\sqrt{2}}}} = \sqrt[12]{2}$